

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'GP_FFCutsFits.mat');

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if ~exist('GP_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(GP_FFCutsFits);
for i = 1:numel(fn)
    selectCuts = fn{i};
    if contains(selectCuts, 'msk')
        continue
    end
    if contains(selectCuts, 'all_fit_par')
        continue
    end

    fprintf('Processing cuts: %s \n', selectCuts);
    %selectCuts= 'Ei400_off200GPmaxFF';
    cuts_list = GP_FFCutsFits.(selectCuts);
    if ~iscell(cuts_list)
        cuts_list = {cuts_list};
    end
    if contains(selectCuts, 'Ei200')
        mi = maps_instrument(200,600, 'S');
    end
    if contains(selectCuts, 'Ei400')
        mi = maps_instrument(401,600, 'S');
    end
    if contains(selectCuts, 'Ei800')
        mi = maps_instrument(786,600, 'S');
    end
    sample=IX_sample(true,[1,0,0],[0,1,0], 'cuboid',[0.04,0.03,0.02]);
    for j=1:numel(cuts_list)
        cuts_list{j} = cuts_list{j}.set_instrument(mi);
        cuts_list{j} = cuts_list{j}.set_sample(sample);
        plot(cuts_list{j}); lz 0 1; keep_figure;
    end
    dir_name = "GP";

    dE_step = 2; %original energy transfer step data were binned to. No point in
going finer
    half_dE = 5; % half width of data binning

    en_mm = cuts_list{1}.data.img_range(:,4);

```

```

en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 170;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
GP_FFCutsFits.(selectCuts) = cuts_list;

GP_FFCutsFits = fit_multicutsFF_along_direction(...
    GP_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
cl = GP_FFCutsFits.(selectCuts);
GP_FFCutsFits.(selectCuts) = [cl{:}];

beep
beep
beep

all_fit_par = GP_FFCutsFits.([selectCuts,'_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure;
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;

%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_GP.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_GP_',selectCuts,'.pdf']);
export(thisFile,targ_file);
close all;
end

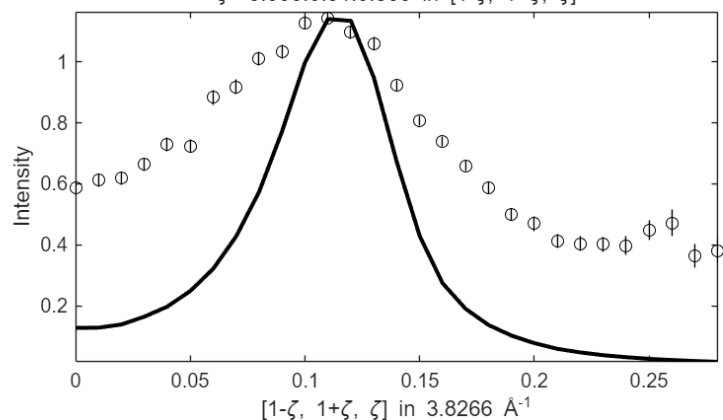
```

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Processing cuts: Ei800_off110GPminFF
*****
**** En = 73±5 going to 253
*****

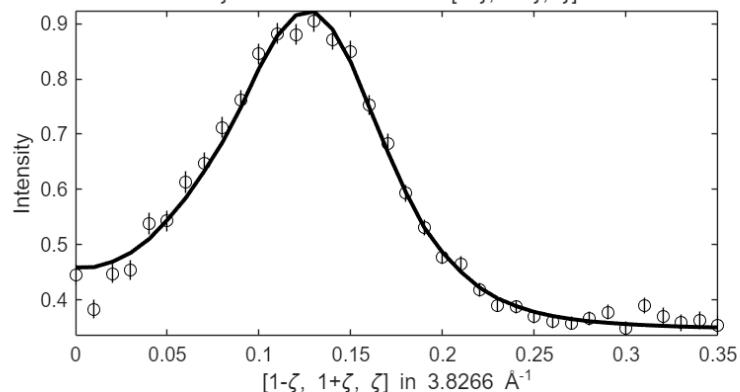
```

Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $39 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

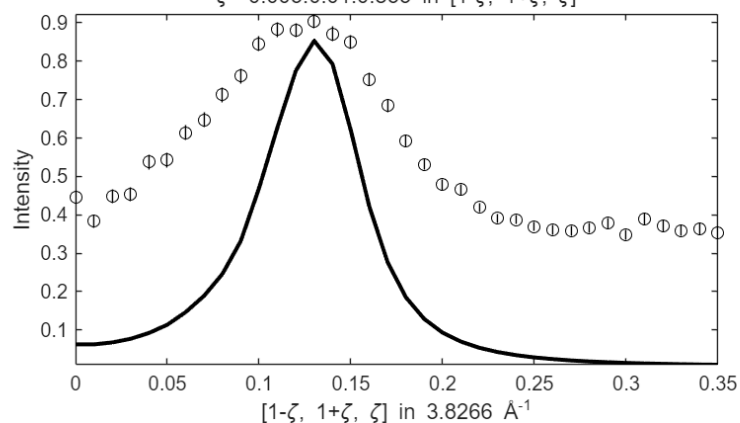


**** En = 54±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.91\pm0.015$; $J_0=25\pm0.24$; $\gamma=16.8159\pm0.245304$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

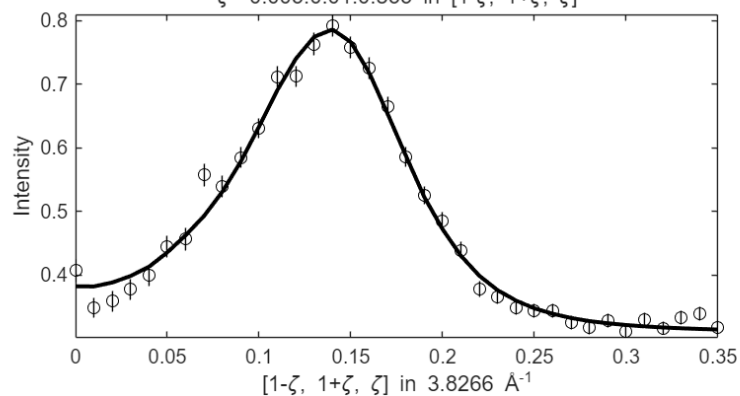


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $49 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

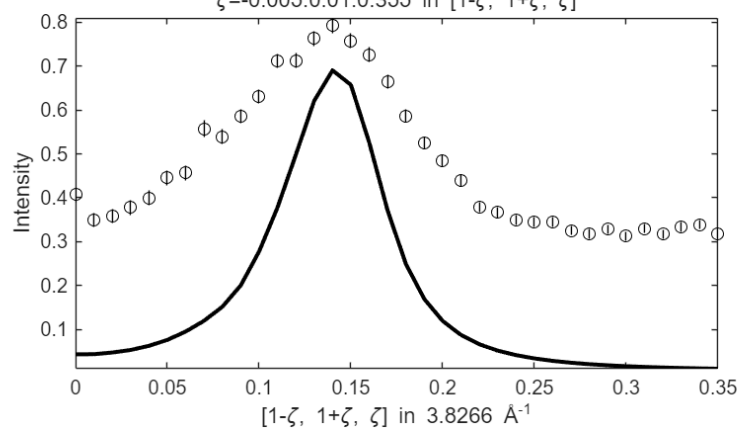


**** En = 64±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.83\pm0.014$; $J0= 27\pm0.23$; $\gamma=19.3688\pm0.26398$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

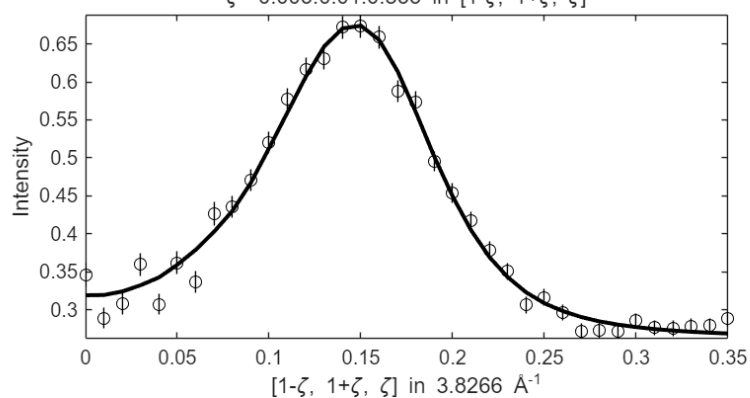


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $59 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

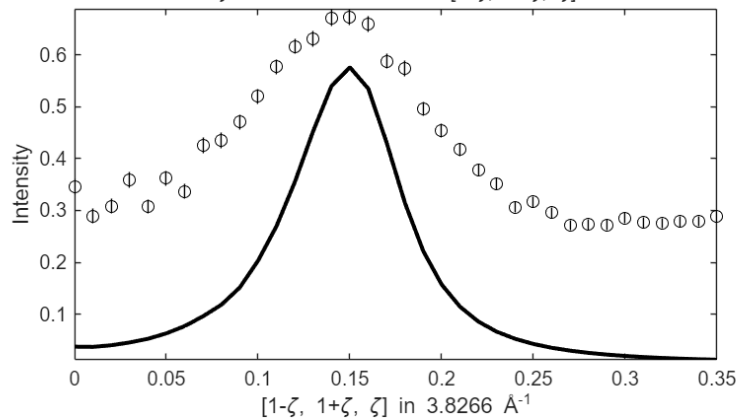


**** En = 74 ± 5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.83\pm0.015$; $J0= 29\pm0.25$; $\gamma=23.7402\pm0.336461$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

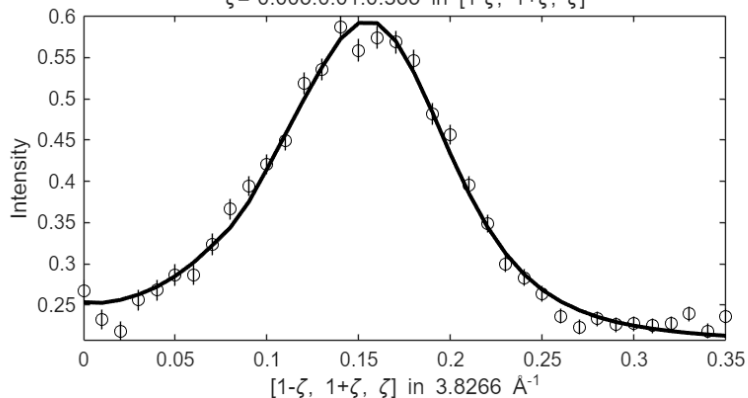


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $69 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

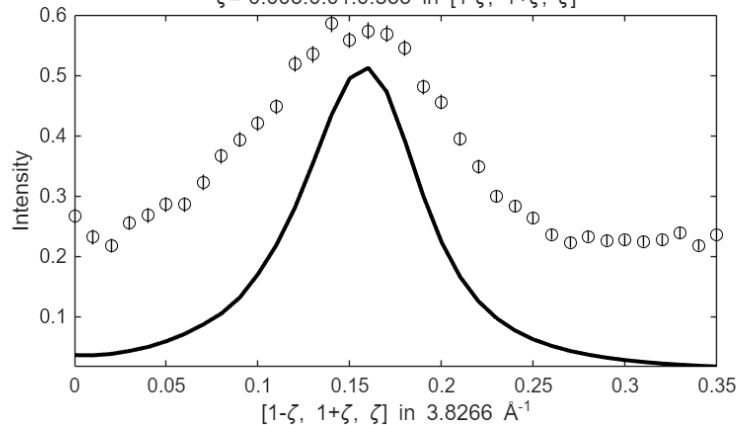


**** En = 84±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92 \pm 0.019$; $J_0 = 30 \pm 0.25$; $\gamma = 30.5252 \pm 0.440487$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

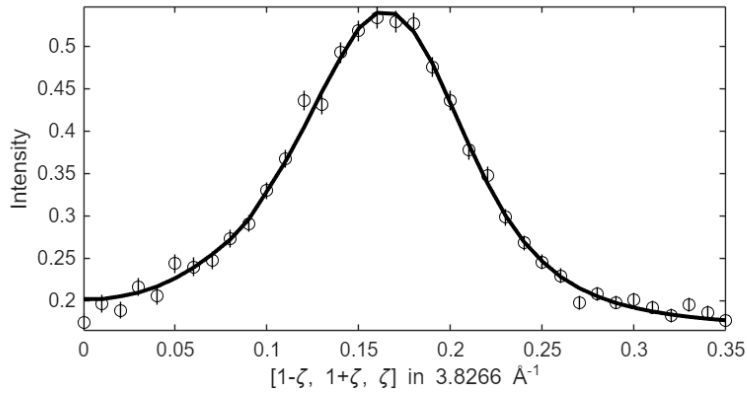


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $79 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

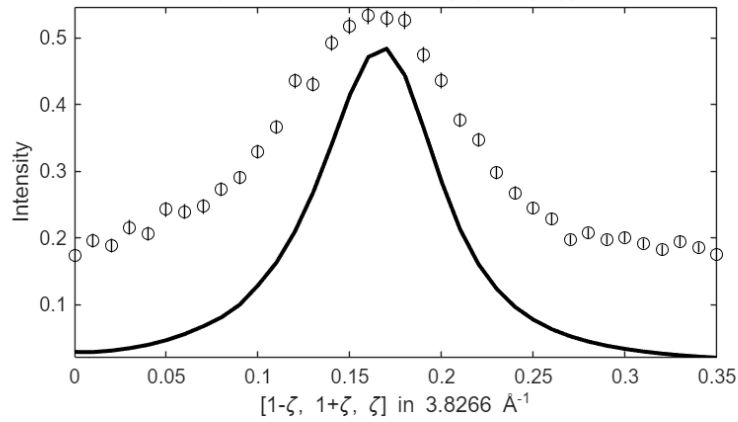


**** En = 94±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.92\pm0.048$; $J_0= 31\pm 0.3$; $\gamma=32.2884\pm2.42507$
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

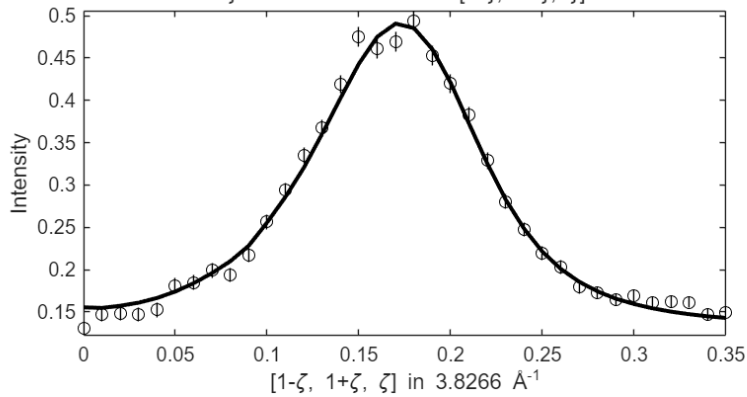


Fe_ei401_align_sym3, w2_400off110.sqw
 $.1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $89 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

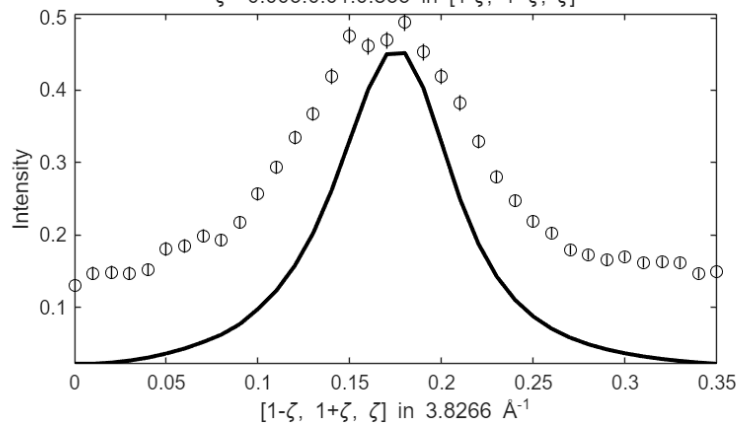


 **** En = 104±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.89\pm0.018$; $J_0= 32\pm0.18$; $\gamma=32.8522\pm0.486961$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E$:
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

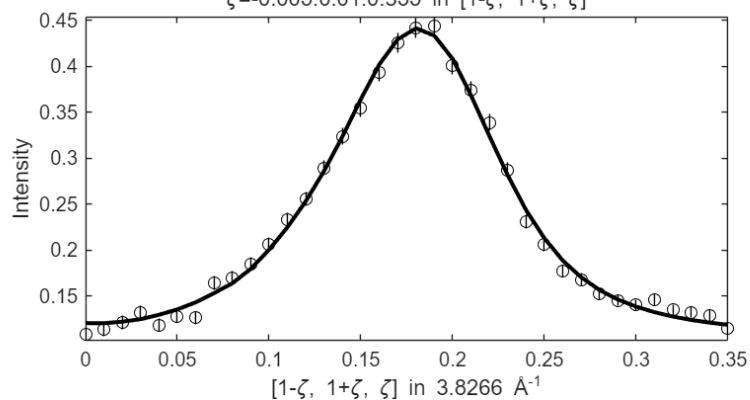


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $99 \leq E \leq 101$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

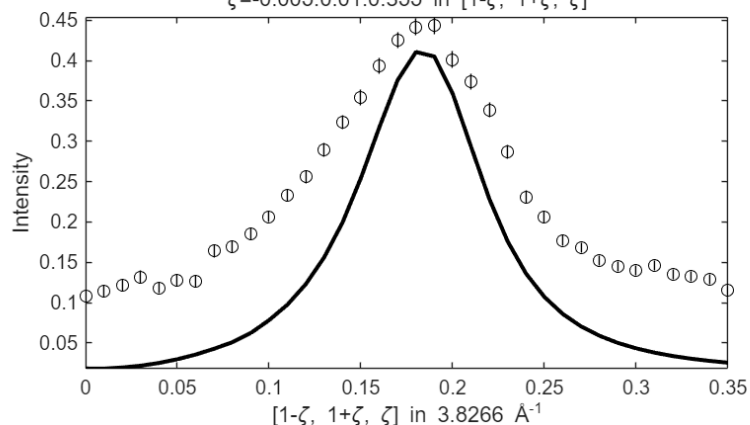


**** En = 114±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.89 \pm 0.017$; $J_0 = 32 \pm 0.14$; $\gamma = 36.35 \pm 0.539992$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E \leq 111$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

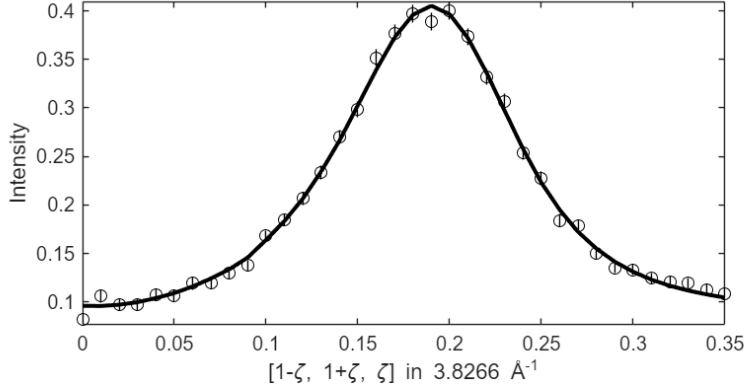


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $109 \leq E \leq 111$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

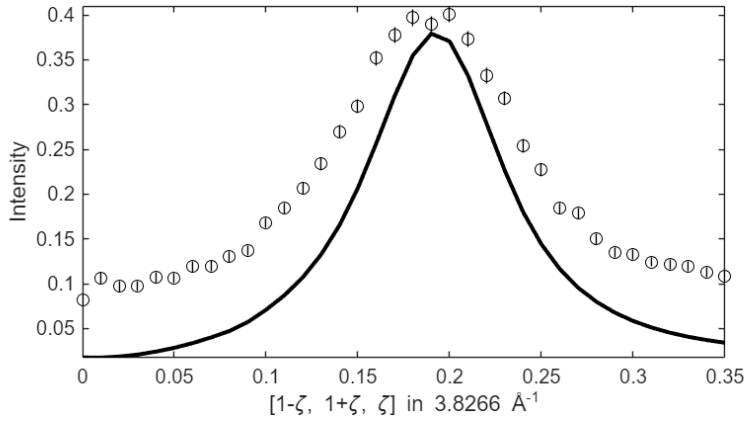


**** En = 124±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.95\pm0.02$; $J0= 33\pm0.11$; $\gamma=43.0912\pm0.758289$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

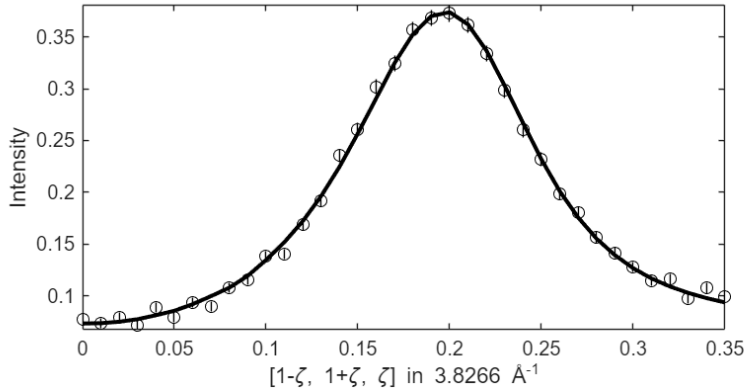


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $119 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

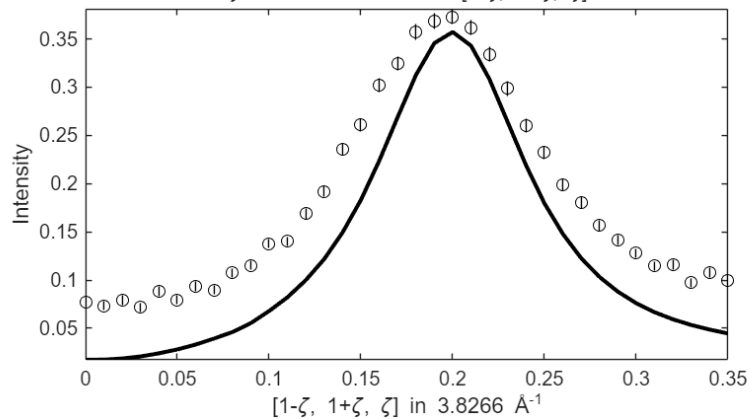


**** En = 134±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S= 1\pm0.029$; $J0= 35\pm0.15$; $\gamma=50.2396\pm1.41398$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta=-0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

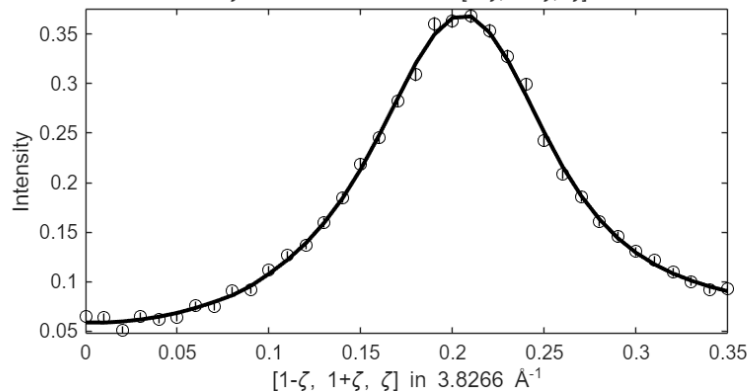


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $129 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

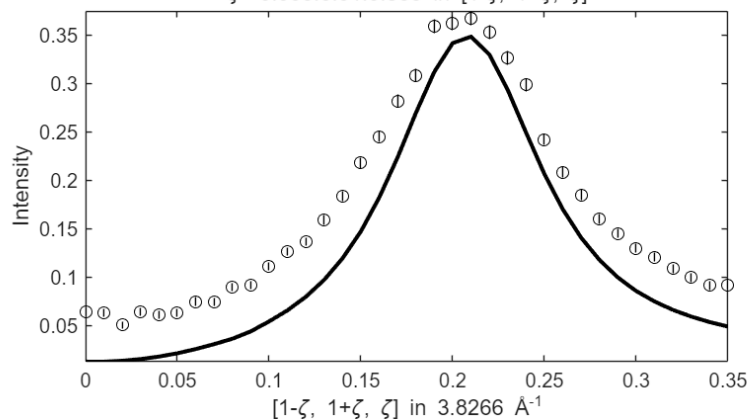


**** En = 144±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.024$; $J_0 = 35 \pm 0.15$; $\gamma = 49.7036 \pm 1.47536$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

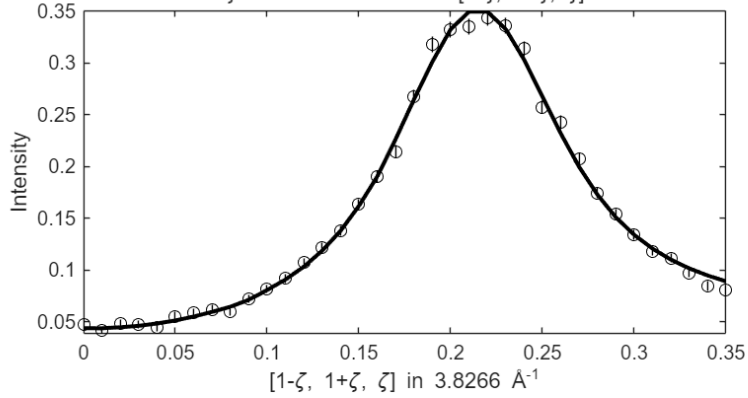


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $139 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

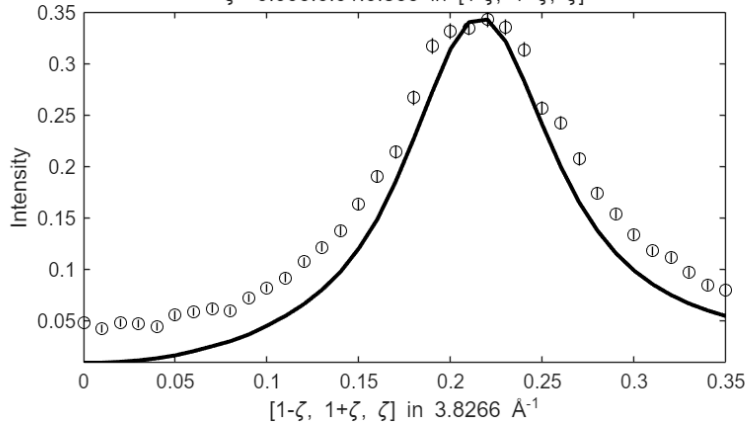


**** En = 154±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.026$; $J_0 = 35 \pm 0.16$; $\gamma = 50.2535 \pm 1.51931$
 $l \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

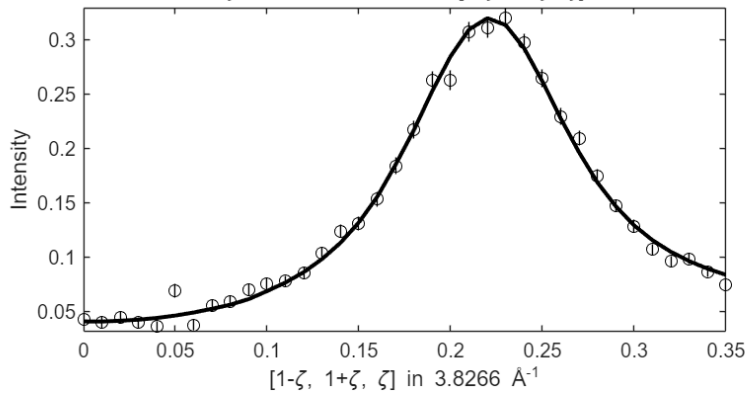


Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 1 \pm 0.026$; $J_0 = 35 \pm 0.16$; $\gamma = 50.2535 \pm 1.51931$
 $l \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $149 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

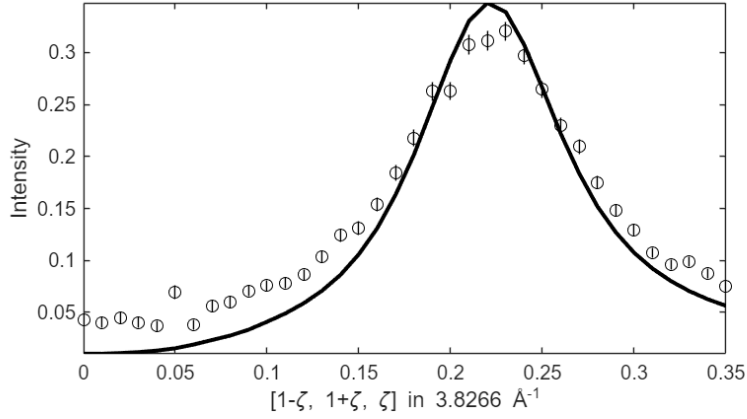


 **** En = 164±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 0.94 \pm 0.036$; $J_0 = 36 \pm 0.23$; $\gamma = 46.5992 \pm 2.13987$
 $l \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

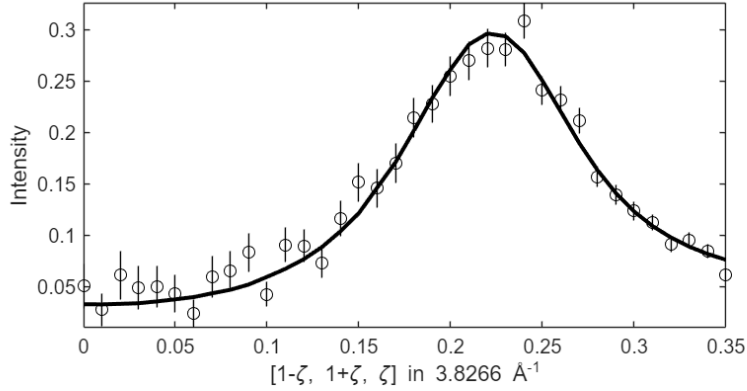


Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $159 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

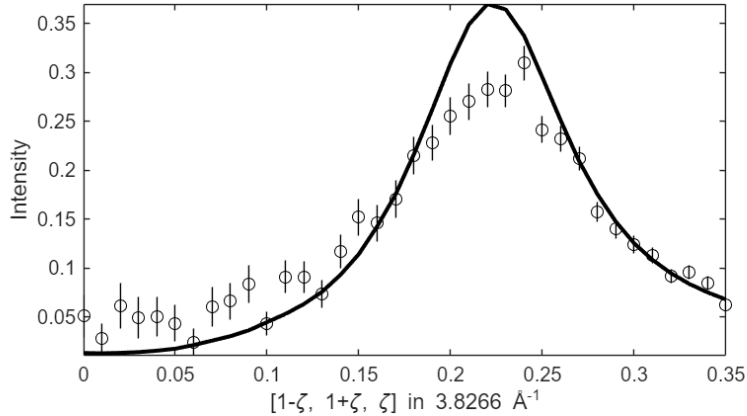


**** En = 174±5 going to 194

Fe_ei401_align_sym3, w2_400off110.sqw
 $S=0.95\pm0.058$; $J_0=35\pm0.41$; $\gamma=45.6768\pm3.55527$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $169 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



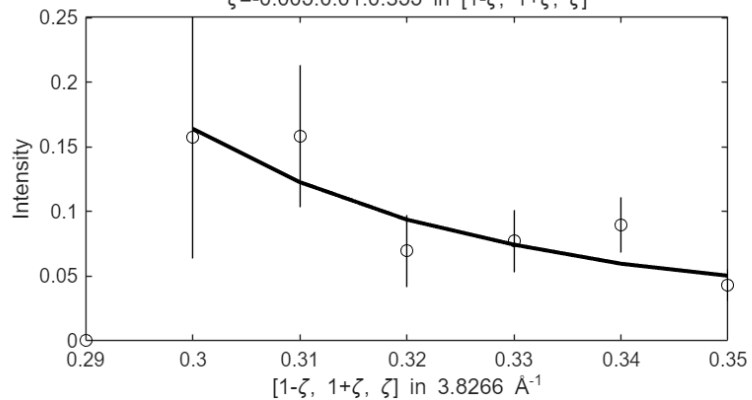
Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $169 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



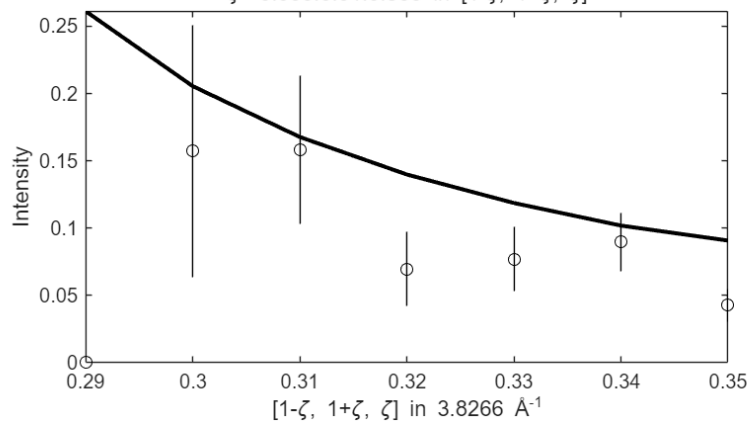
**** En = 184±5 going to 194

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.7778

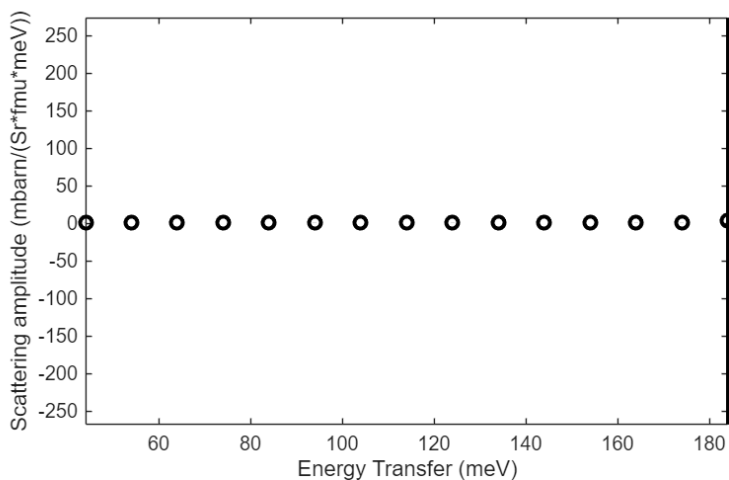
Fe_ei401_align_sym3, w2_400off110.sqw
 $S = 3.2 \pm 2.7 \times 10^2$; $J_0 = 32 \pm 33$; $\gamma = 9.05621 \pm 736.002$
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $179 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$

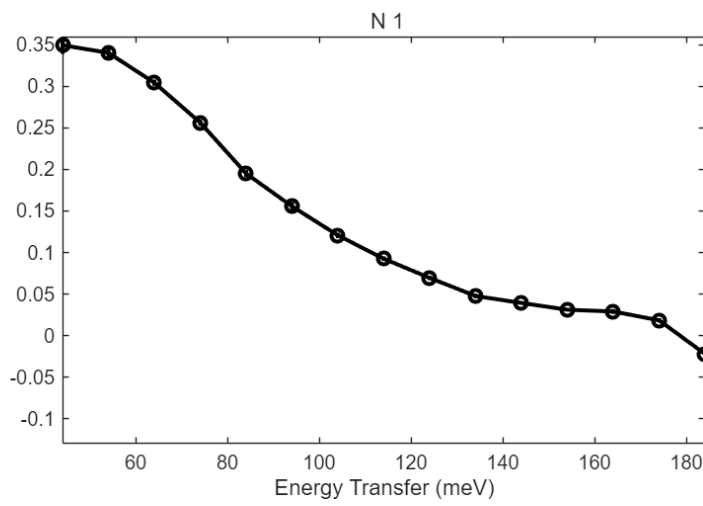
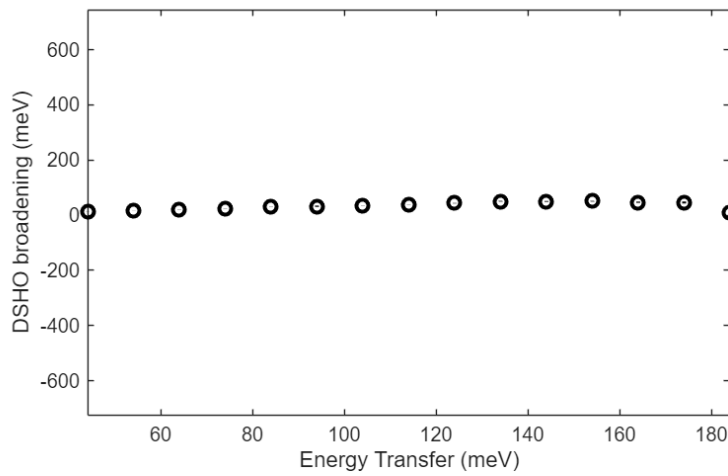
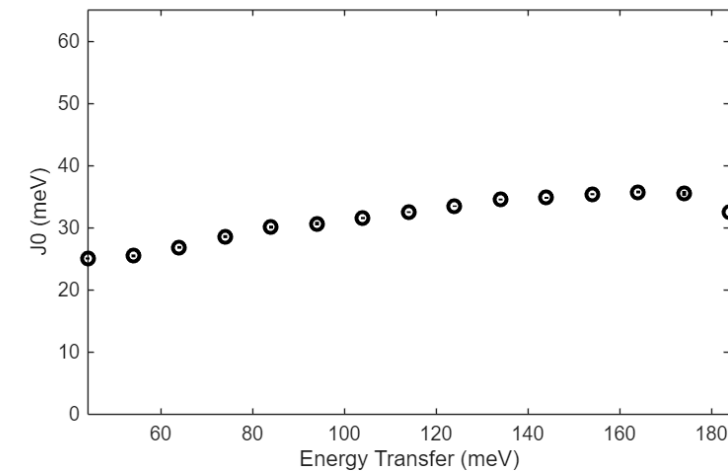


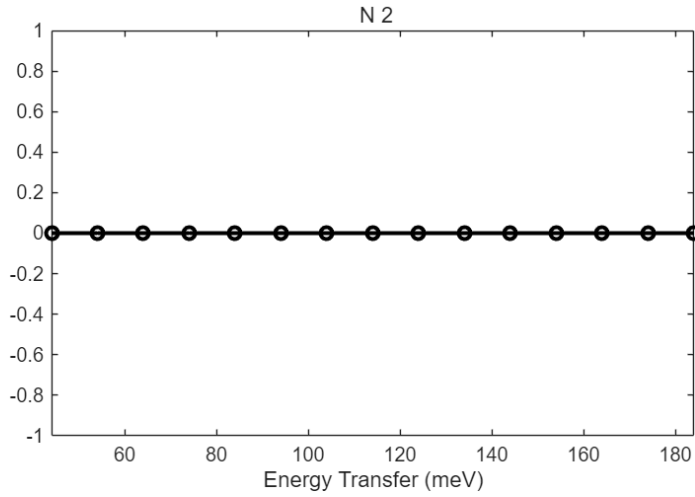
Fe_ei401_align_sym3, w2_400off110.sqw
 $1 \leq \xi \leq 0.1$ in $[1-\xi, 1-\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[1+0.5\eta, 1-0.5\eta, \eta]$, $179 \leq E$
 $\zeta = -0.005:0.01:0.355$ in $[1-\zeta, 1+\zeta, \zeta]$



 **** En = 194±5 going to 194



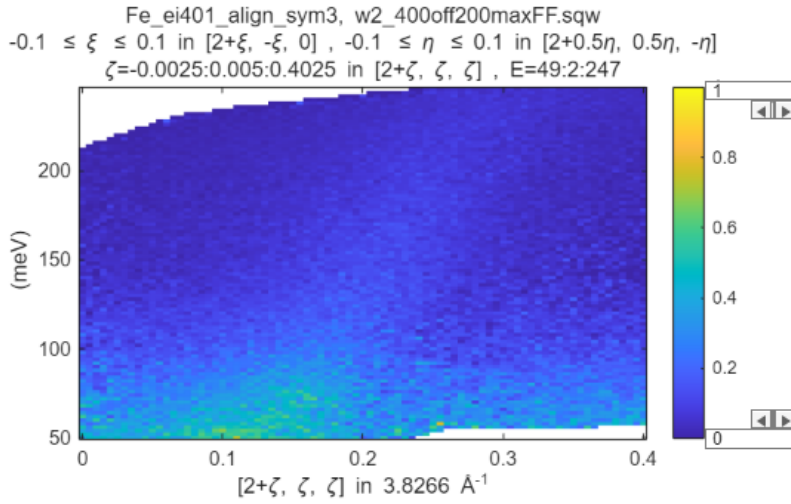




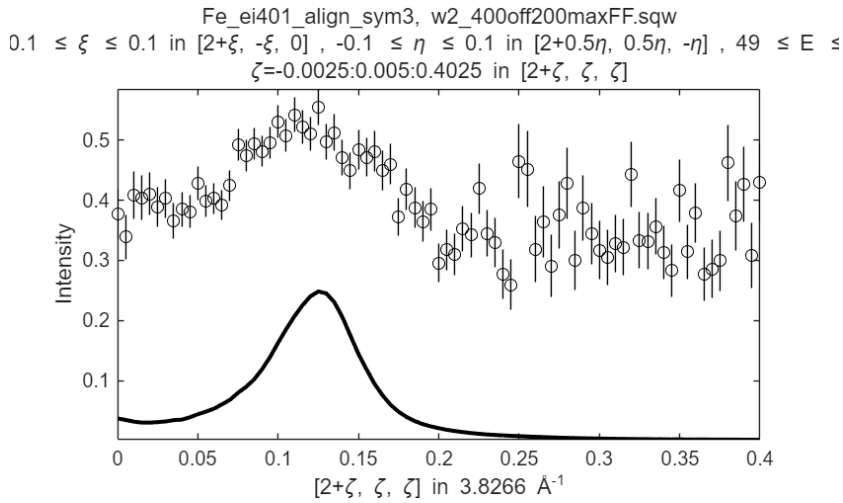
Processing cuts: Ei400_off110GPmaxFF

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*****
**** En = 44±5 going to 164
*****
*****
**** En = 54±5 going to 164
*****
*****
**** En = 64±5 going to 164
*****
*****
**** En = 74±5 going to 164
*****
*****
**** En = 84±5 going to 164
*****
*****
**** En = 94±5 going to 164
*****
*****
**** En = 104±5 going to 164
*****
*****
**** En = 114±5 going to 164
*****
*****
**** En = 124±5 going to 164
*****
*****
**** En = 134±5 going to 164
*****
*****
**** En = 144±5 going to 164
*****
*****
**** En = 154±5 going to 164
*****
*****
**** En = 164±5 going to 164
*****
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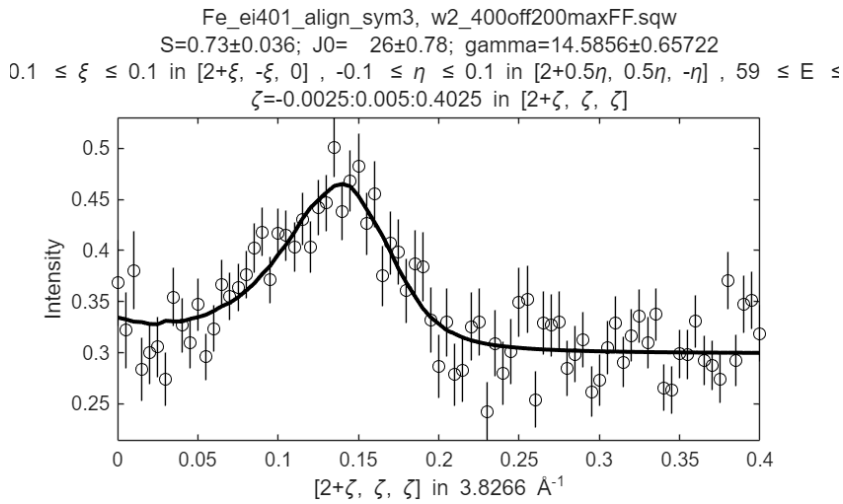
Processing cuts: Ei400_off200GPmaxFF



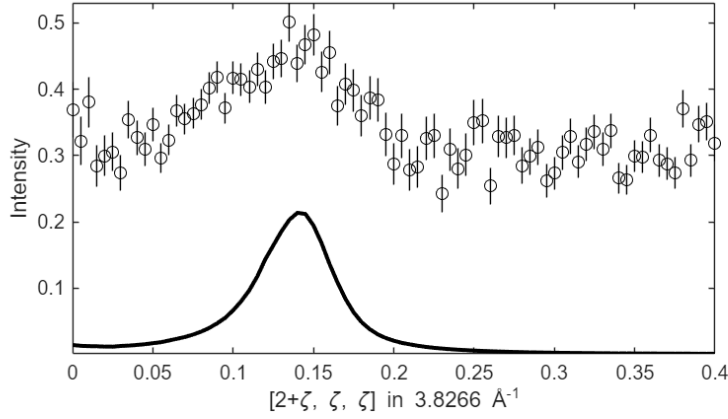
 **** En = 54±5 going to 234



 **** En = 64±5 going to 234

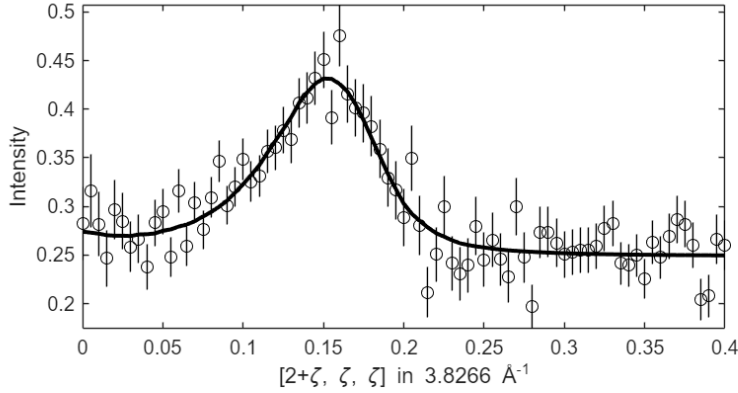


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $59 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

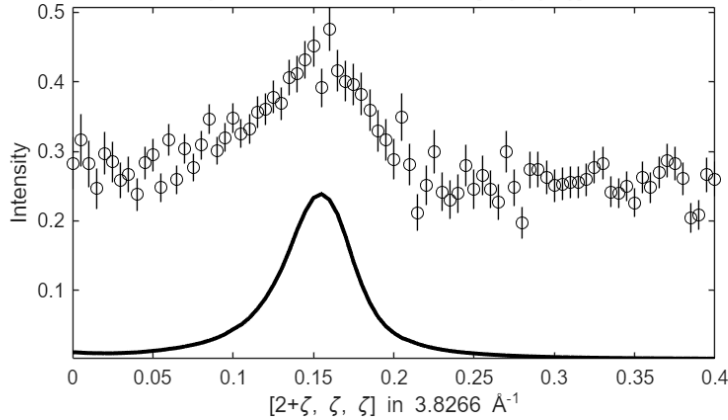


**** En = 74±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.83 \pm 0.033$; $J_0 = 26 \pm 0.53$; $\gamma = 14.48 \pm 0.514321$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $69 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

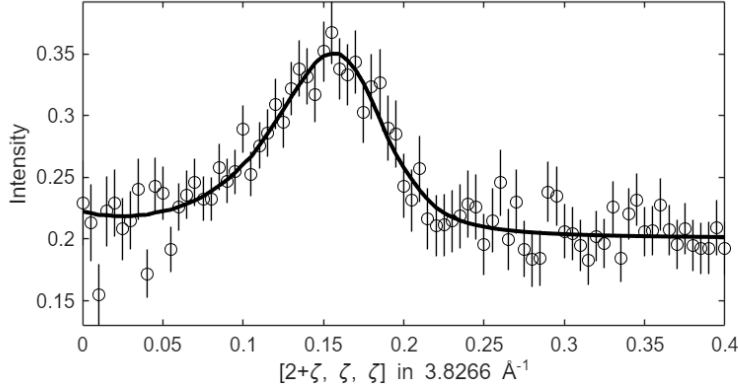


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $69 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

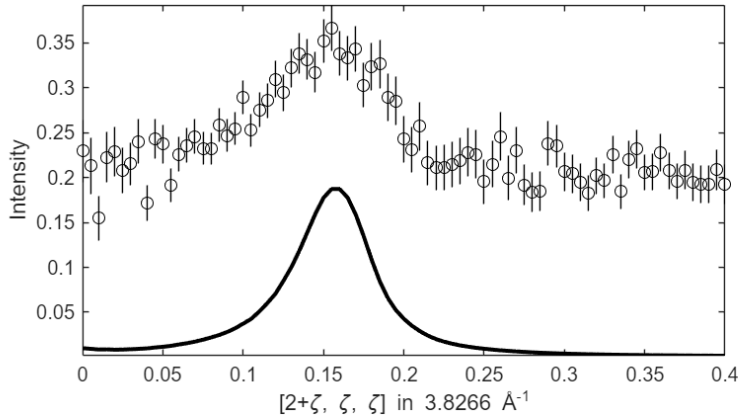


**** En = 84±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.81\pm0.028$; $J_0=29\pm0.52$; $\gamma=18.7132\pm0.543831$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $79 \leq E \leq 81$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

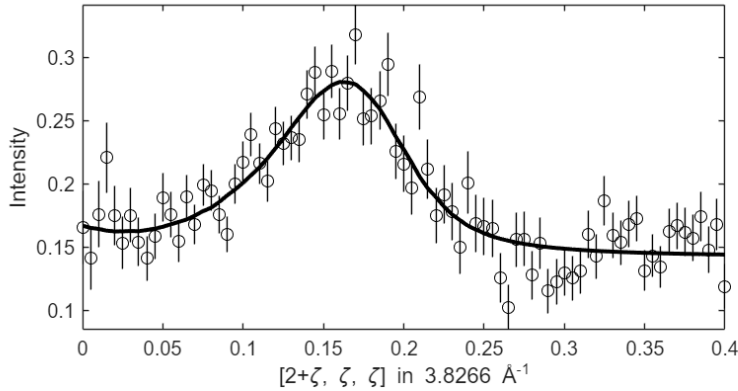


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.81\pm0.028$; $J_0=29\pm0.52$; $\gamma=18.7132\pm0.543831$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $79 \leq E \leq 81$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

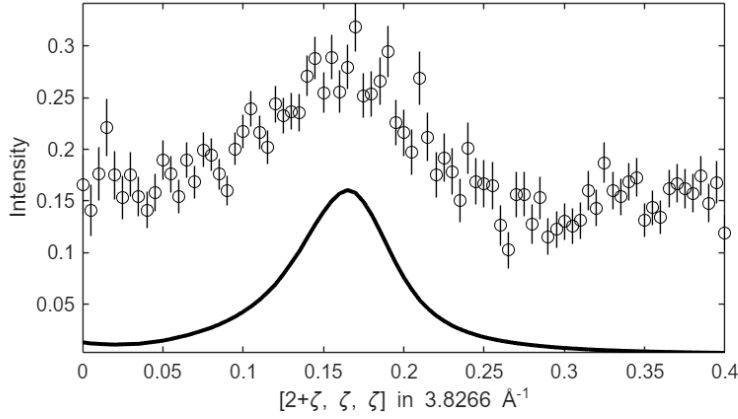


 **** En = 94±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=1.1\pm0.16$; $J_0=31\pm0.85$; $\gamma=30.8005\pm7.02824$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $89 \leq E \leq 91$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

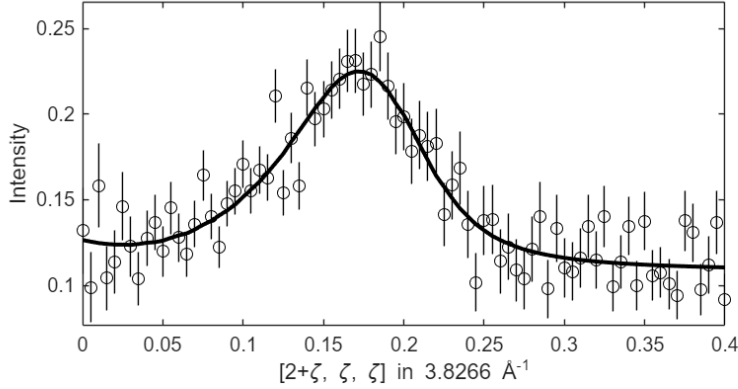


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $89 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

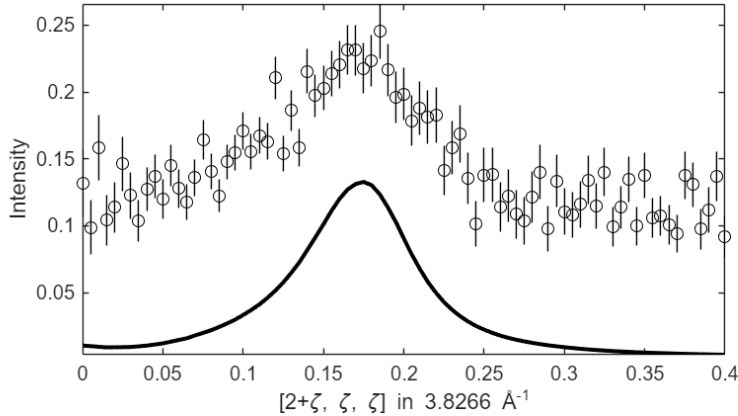


**** En = 104±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1 \pm 0.14$; $J_0 = 32 \pm 0.77$; $\gamma = 35.8417 \pm 7.18465$
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $99 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

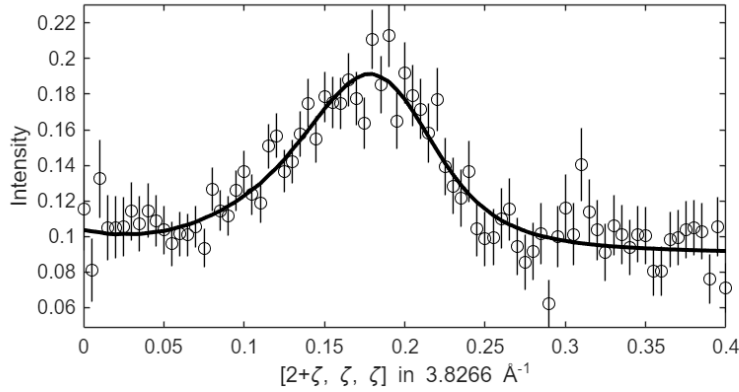


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $0.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $99 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

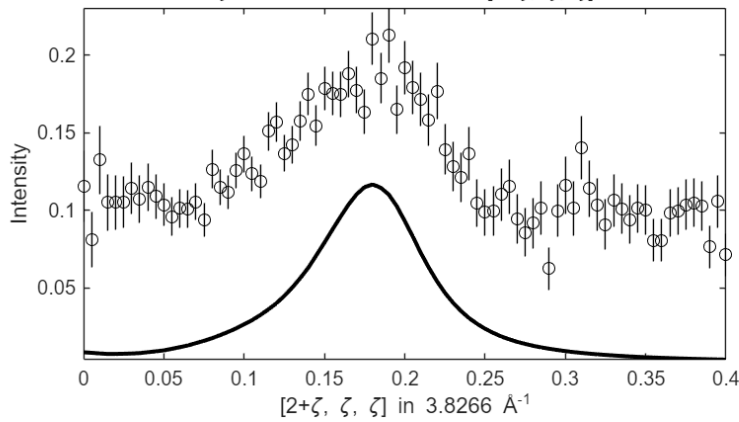


**** En = 114±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.93\pm0.035$; $J_0=33\pm0.58$; $\gamma=37.763\pm0.895301$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $109 \leq E \leq 119$ eV
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

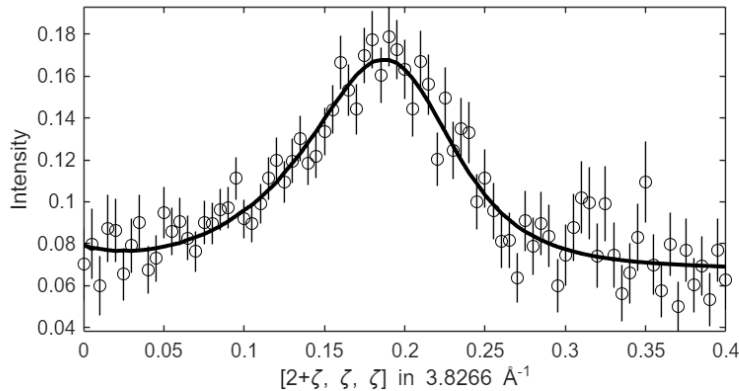


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.93\pm0.035$; $J_0=33\pm0.58$; $\gamma=37.763\pm0.895301$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $109 \leq E \leq 119$ eV
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

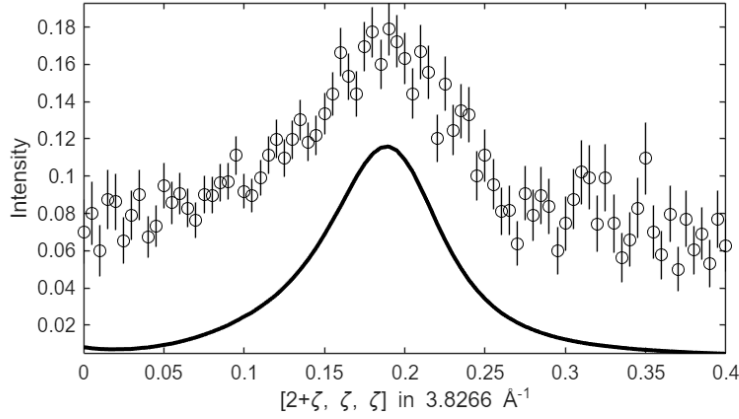


 **** En = 124±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=1\pm0.037$; $J_0=34\pm0.46$; $\gamma=41.8755\pm0.808517$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $119 \leq E \leq 129$ eV
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

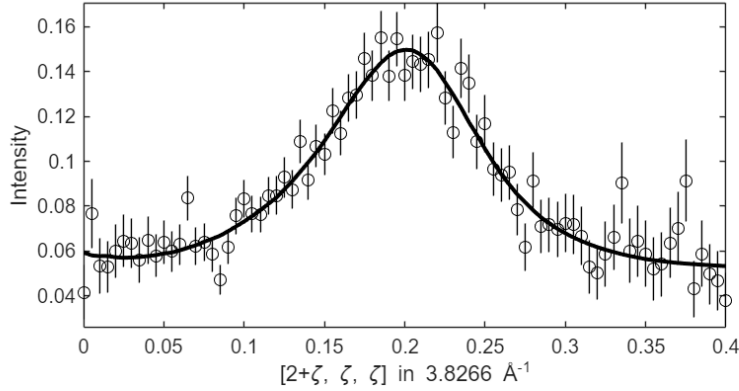


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $119 \leq E \leq 129$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

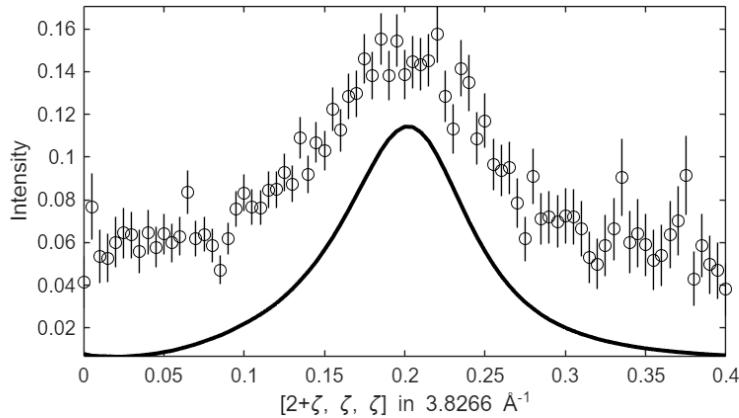


 **** En = 134±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.2 \pm 0.12$; $J_0 = 34 \pm 0.5$; $\gamma = 46.5788 \pm 5.82878$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $129 \leq E \leq 139$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

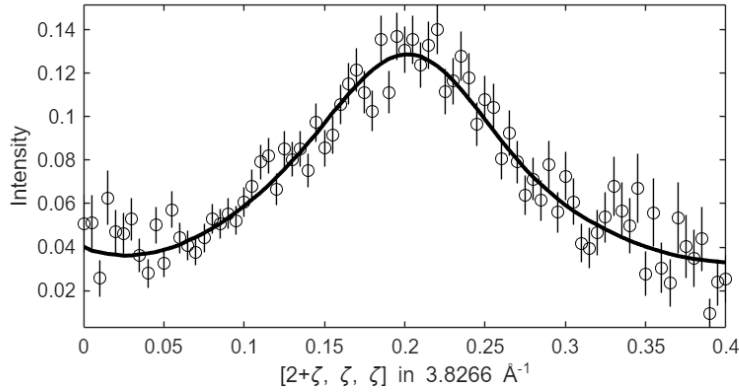


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $129 \leq E \leq 139$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

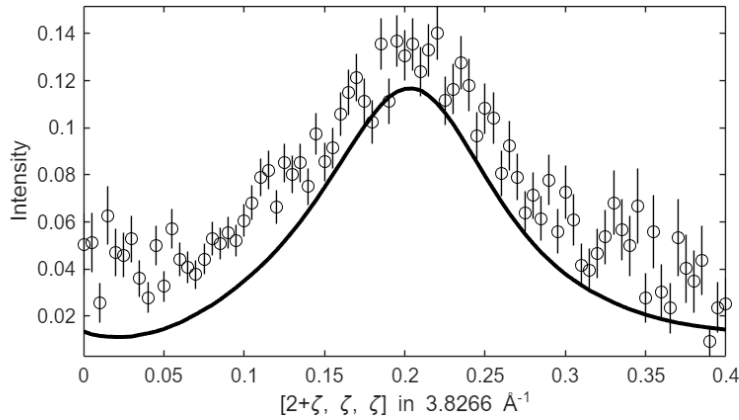


 **** En = 144±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.9 \pm 0.2$; $J_0 = 37 \pm 0.65$; $\gamma = 79.9067 \pm 7.10633$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $139 \leq E \leq 149$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

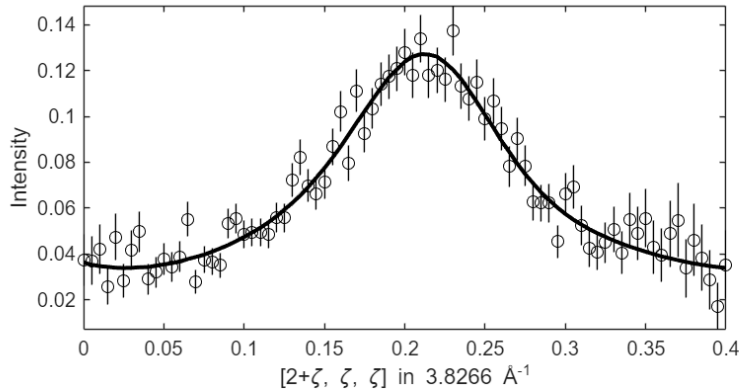


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.9 \pm 0.2$; $J_0 = 37 \pm 0.65$; $\gamma = 79.9067 \pm 7.10633$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $139 \leq E \leq 149$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

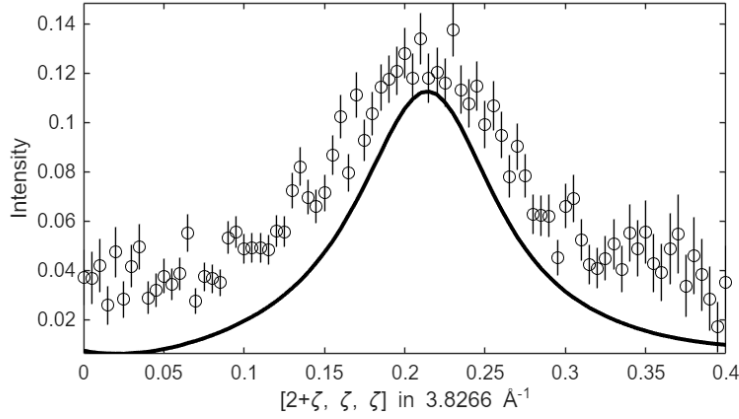


 **** En = 154±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.072$; $J_0 = 35 \pm 0.31$; $\gamma = 53.138 \pm 2.14837$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $149 \leq E \leq 159$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

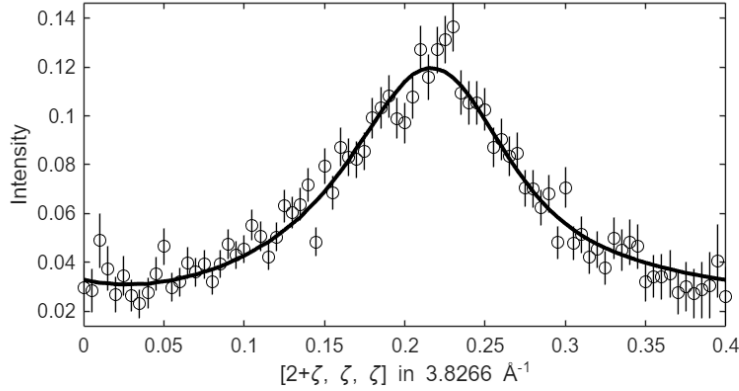


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $149 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

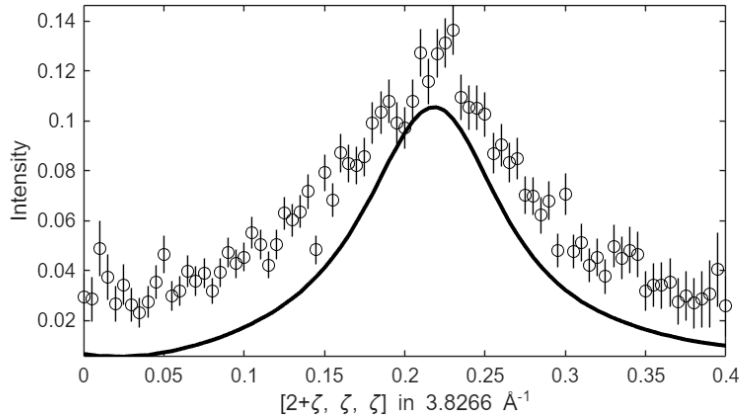


**** En = 164±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.067$; $J_0 = 37 \pm 0.28$; $\gamma = 55.1241 \pm 2.13363$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $159 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

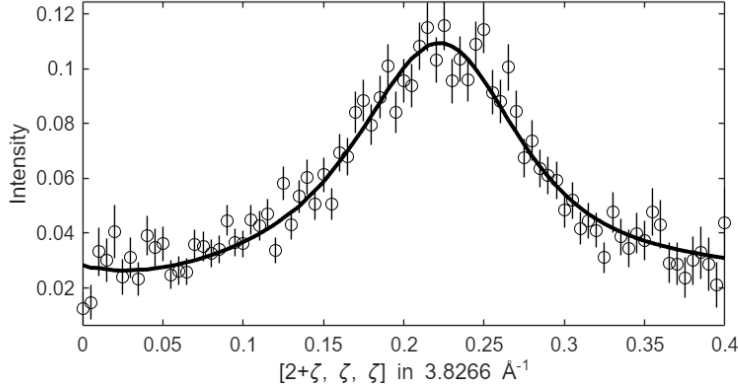


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $159 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

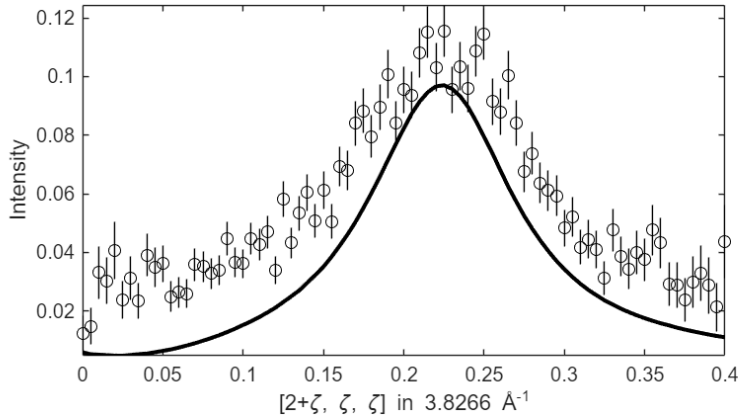


**** En = 174±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.11$; $J_0 = 37 \pm 0.45$; $\gamma = 57.7493 \pm 5.7971$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $169 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

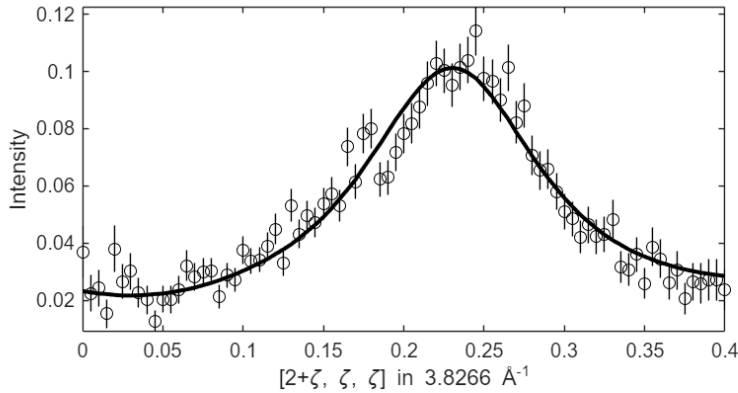


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.11$; $J_0 = 37 \pm 0.45$; $\gamma = 57.7493 \pm 5.7971$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $169 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

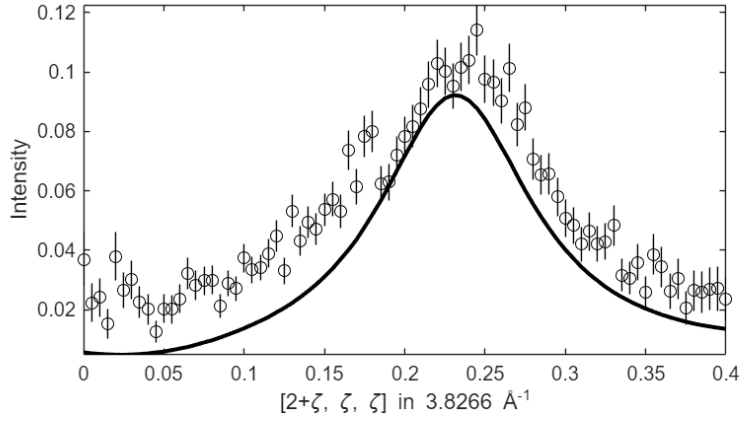


 **** En = 184±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.095$; $J_0 = 38 \pm 0.35$; $\gamma = 61.3767 \pm 4.09929$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $179 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

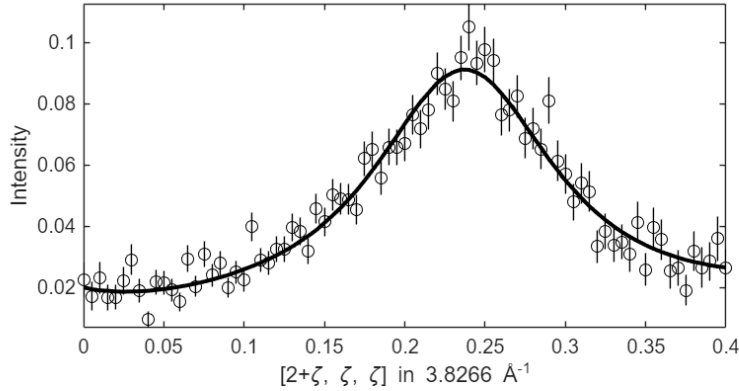


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $179 \leq E \leq 189$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

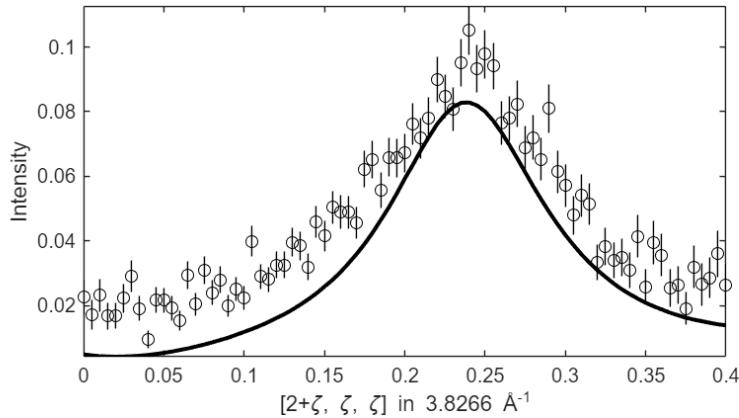


 **** En = 194±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.2 \pm 0.093$; $J_0 = 38 \pm 0.41$; $\gamma = 63.3978 \pm 5.52731$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $189 \leq E \leq 199$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

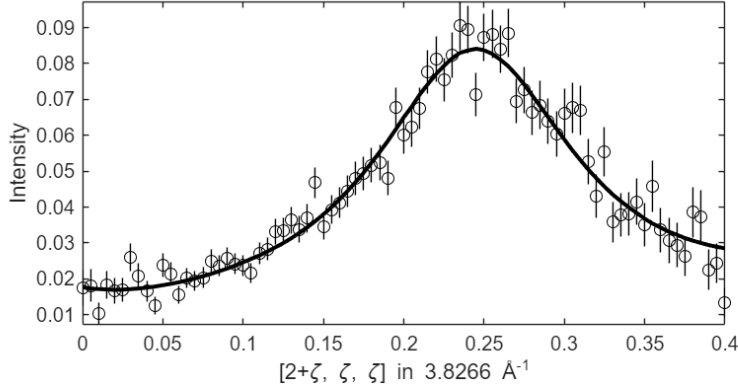


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $189 \leq E \leq 199$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

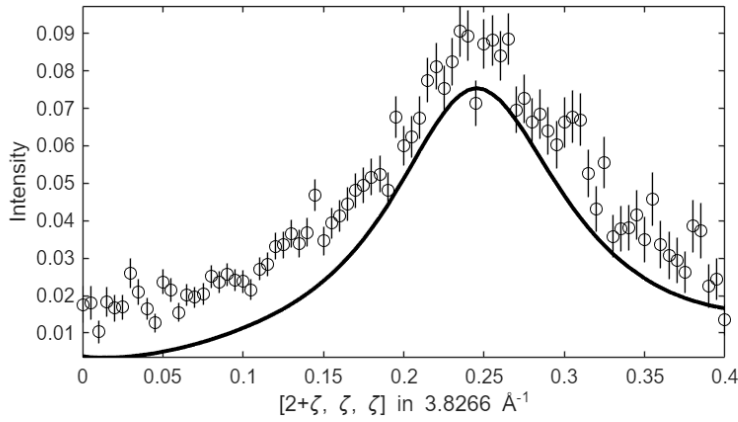


 **** En = 204±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.084$; $J_0 = 39 \pm 0.39$; $\gamma = 70.6115 \pm 5.42936$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $199 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

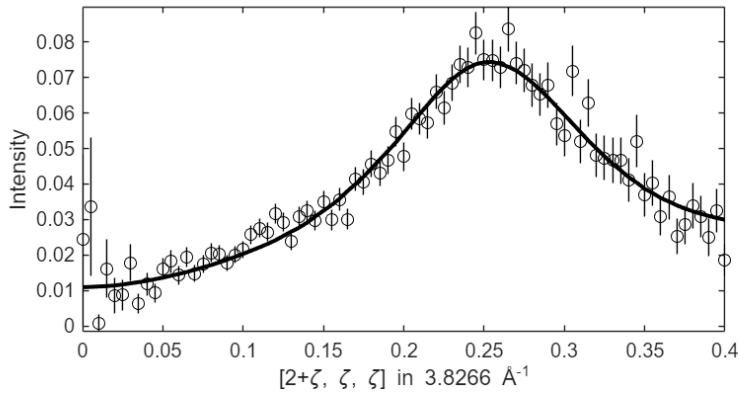


Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.084$; $J_0 = 39 \pm 0.39$; $\gamma = 70.6115 \pm 5.42936$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $199 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$

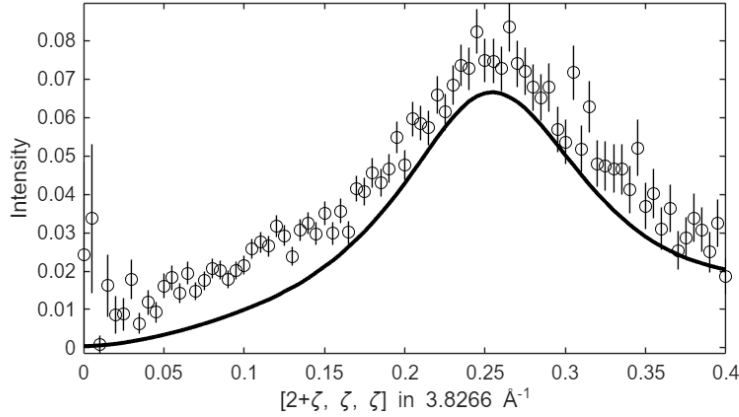


 **** En = 214±5 going to 234

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.3 \pm 0.08$; $J_0 = 39 \pm 0.4$; $\gamma = 78.745 \pm 5.70504$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $209 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



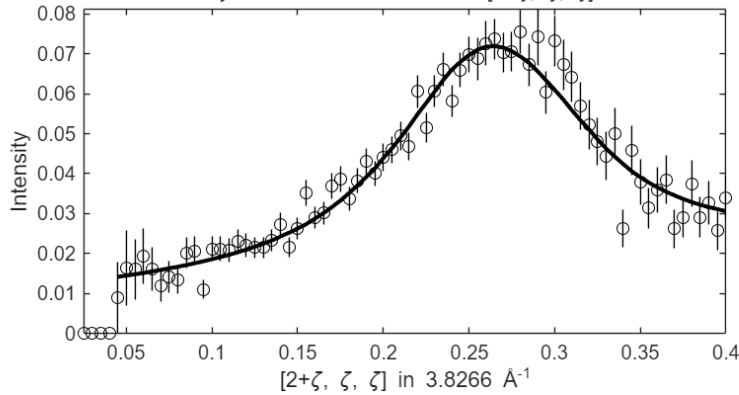
Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $209 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



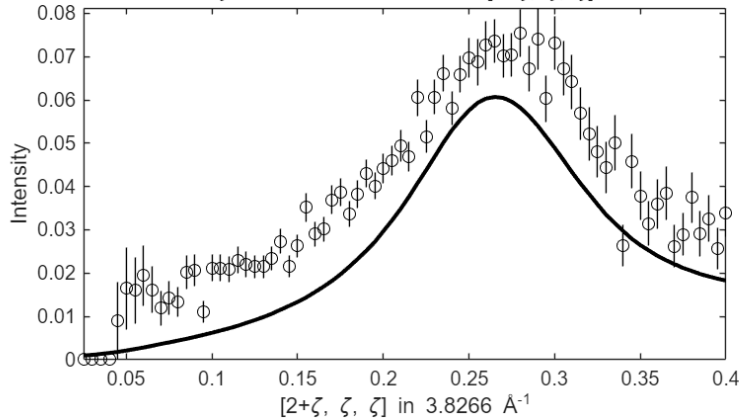
**** En = 224±5 going to 234

Warning: Dataset:4 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 4.9383

Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S=0.99\pm0.073$; $J0=39\pm0.31$; $\gamma=64.8753\pm4.91672$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $219 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



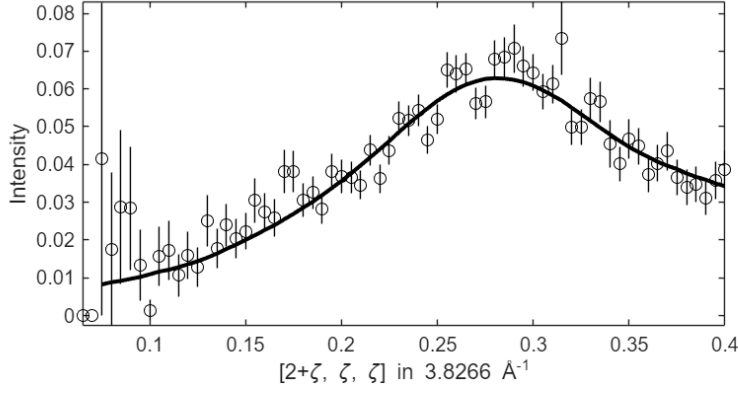
Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $219 \leq E \leq$
 $\zeta=-0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



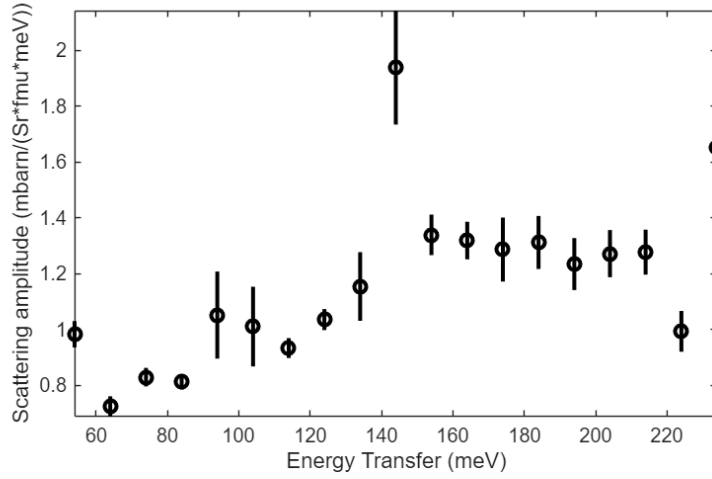
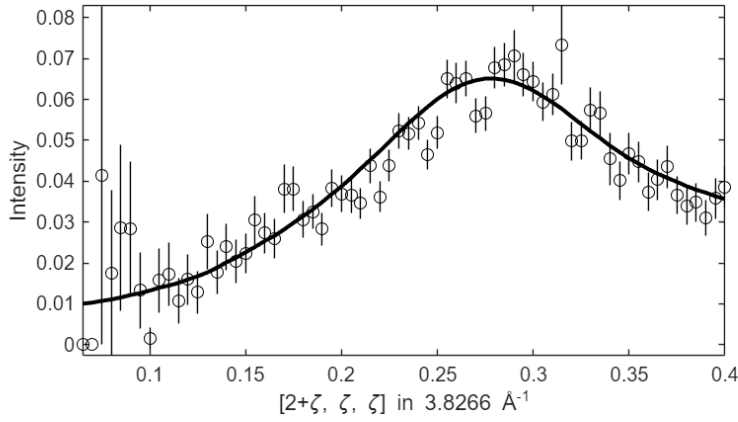
**** En = 234±5 going to 234

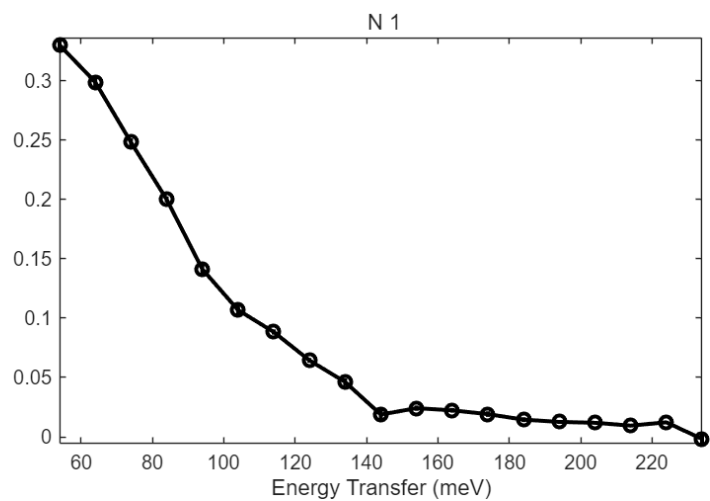
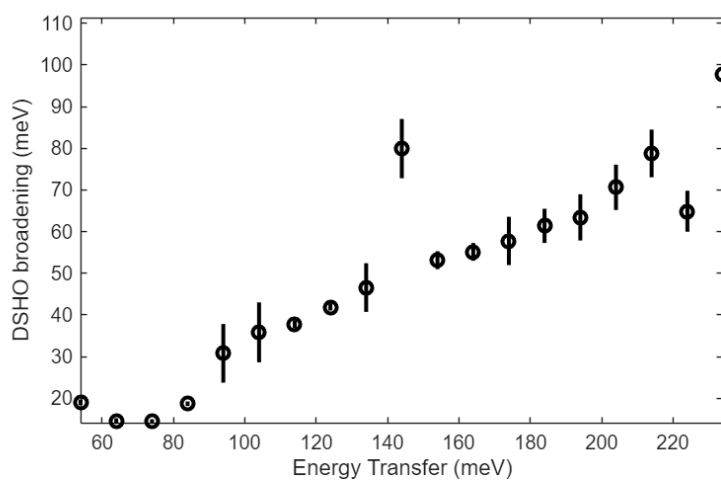
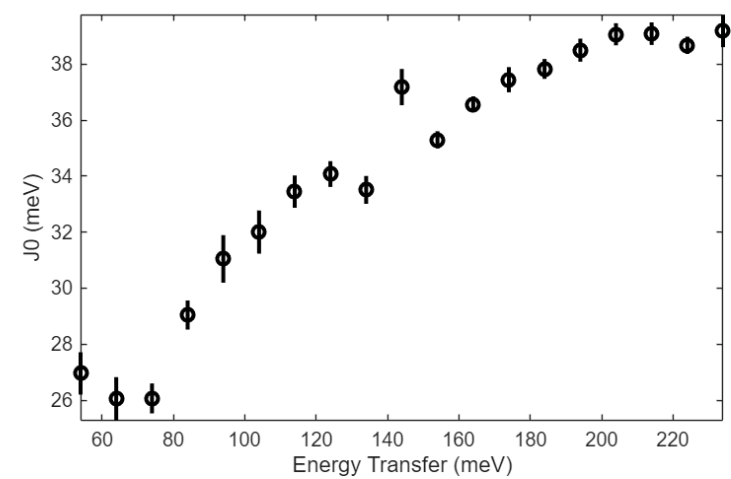
Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 2.4691

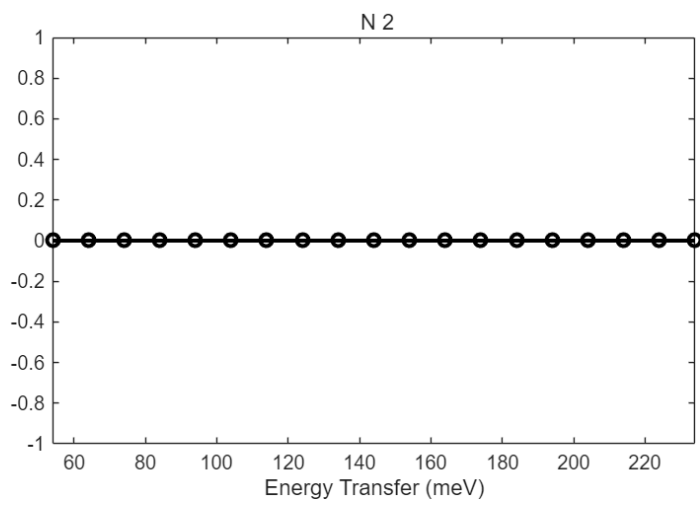
Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $S = 1.7 \pm 0.28$; $J_0 = 39 \pm 0.58$; $\gamma = 97.6814 \pm 13.6481$
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $229 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$



Fe_ei401_align_sym3, w2_400off200maxFF.sqw
 $.1 \leq \xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2+0.5\eta, 0.5\eta, -\eta]$, $229 \leq E \leq$
 $\zeta = -0.0025:0.005:0.4025$ in $[2+\zeta, \zeta, \zeta]$







```
fit_src_struct = GP_FFCutsFits;  
save(Cutfitfile,"fit_src_struct",-v7.3')
```